Started on	Sunday, 25 April 2021, 12:54 PM
State	Finished
Completed on	Sunday, 25 April 2021, 2:24 PM
Time taken	1 hour 30 mins
Grade	**14.00** out of 15.00 (**93**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Consider a reverse-biased long GaAs pn junction diode ($n_i=10^6$ cm^{-3}) at T=300K (V_T=26 mV). Assume that a reverse-bias voltage of -1V is applied. Assume parameters of $N_A=N_D=10^{15}$ cm^{-3}, $D_p=6$cm^2/s, $D_n=400$cm^2/s, and $\tau_n=\tau_p=10^{-8}$s.

The current density (in fA/cm^2) in the diode is .....

$J = qn_i^2\left(\frac{D_P}{N_D L_P} + \frac{D_n}{L_n N_A}\right)$

Answer: -0.0359 ✔

The correct answer is: 0.03591

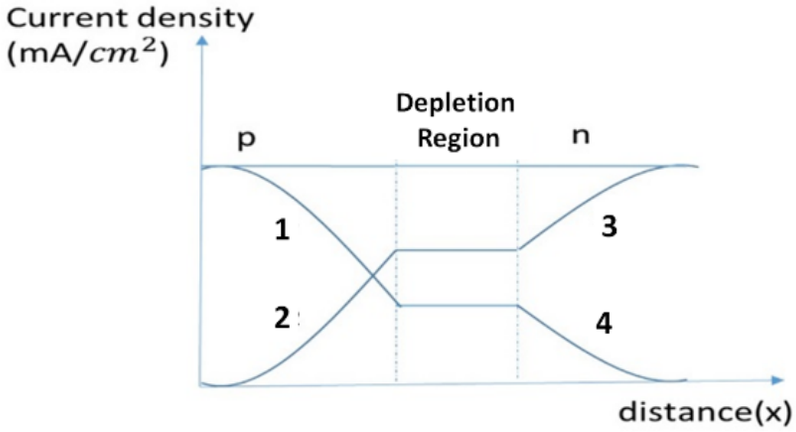
Question **2**

Correct

Mark 1.00 out of 1.00

(Negative mark of -0.25 will be awarded for the wrong answer)

The current density versus distance (spatial map of current) for a forward-biased p-n junction diode is shown in the graph. Assume the electron and hole have equal mobility and lifetime.



Select the most suitable option.

Select one:

- The p side is doped more than the n side
- The n side is doped more than the p side ✔
- Both sides are equally doped
- None of the above

Your answer is correct.

The correct answer is: The n side is doped more than the p side

Question **3**

Incorrect

Mark 0.00 out of 1.00

Consider two p^+n diodes D1 and D2 with a width of the neutral n-region W_{n1} = 40 μm and W_{n2} = 0.1 μm respectively and are identical in all other aspects.

Assuming the diffusion length of holes L_p =2 μm, the ratio of current in D2 to that in D1 at a voltage 0.5 V is I_{D2}/I_{D1}= ....

Answer: 200 20 ✖

The correct answer is: 20

Question **4**

Correct

Mark 1.00 out of 1.00

A p^+n diode under reverse bias shows a current of 2 μA when the probability P of creating an electron-hole pair by undergoing ionizing collision with lattice is zero.

$I = MI_0 \quad M = \frac{1}{1-P} \Rightarrow I = 100\mu A$

The current through the diode under an increased reverse bias when the value of P is found to be 0.98, is ... μA.

Answer: 100 ✔

The correct answer is: 100

Question **5**

Correct

Mark 1.00 out of 1.00

Consider an ideal long p-n junction diode with minority carrier lifetimes $\tau_p=\tau_n=0.1$ μs and the mobilities of electrons and holes are related as $\mu_n=4\mu_p$.

In order to ensure that 90% of the diode current in the depletion region is carried by electrons, the required value of N_A/N_D is

Answer: 0.2222 ✔

$4 \leftarrow \frac{D_n L_p N_D}{D_p L_n N_A} = 9$

$\downarrow$

$1/2$

The correct answer is: 0.22

Question **6**

Correct

Mark 1.00 out of 1.00

A semiconductor sample is illuminated with monochromatic light for which the absorption coefficient of the semiconductor material is 1×10^4 /cm.

The fraction of the incident light absorbed between 0.25 μm and 2 μm from the surface is .....

Answer: 0.643 ✔

$e^{-0.25} - e^{-2}$

The correct answer is: 0.643

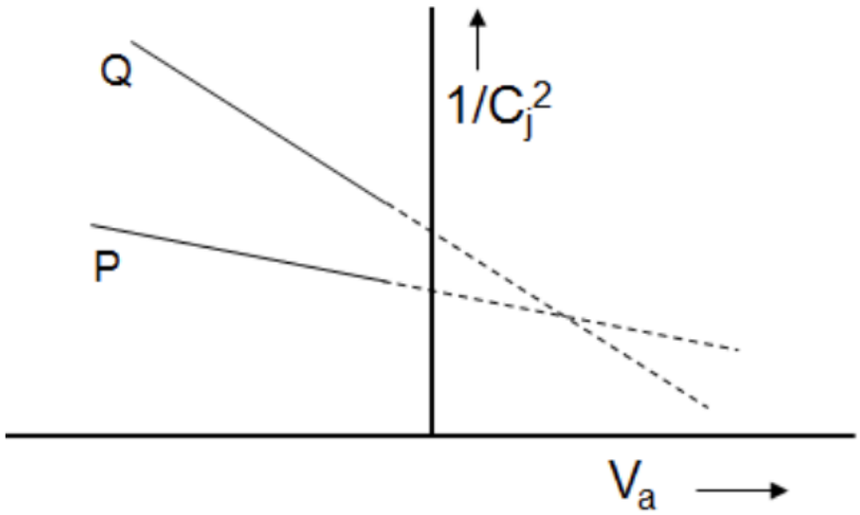
Question **7**

Correct

Mark 1.00 out of 1.00

(Negative mark of -0.25 will be awarded for wrong answer)

P and Q in the figure refer to applied voltage (V_a) dependent inverse square of junction capacitance ($1/C_j^2$) plots corresponding to two abrupt p^+n junction diodes at T=300 K. These two diodes are identical except for their cross-sectional area and n-region doping concentrations. If the cross-sectional area and doping of the n-region corresponding to Q are A_Q=4 μm^2 and $N_{DQ}=10^{14}/cm^3$, respectively,



Select the most suitable option.

Select one:

- Cross-sectional area and doping in the n-region corresponding to P are A_P=2 μm^2 and $N_{DP}=2x10^{13}/cm^3$, respectively.
- Cross-sectional area and doping in the n-region corresponding to P are A_P=2 μm^2 and $N_{DP}=5x10^{13}/cm^3$, respectively.
- Cross-sectional area and doping in the n-region corresponding to P are A_P=5 μm^2 and $N_{DP}=5x10^{14}/cm^3$, respectively. ✔
- Cross-sectional area and doping in the n-region corresponding to P are A_P=5 μm^2 and $N_{DP}=2x10^{13}/cm^3$, respectively.

Your answer is correct.

The correct answer is: Cross-sectional area and doping in the n-region corresponding to P are A_P=5 μm^2 and $N_{DP}=5x10^{14}/cm^3$, respectively.

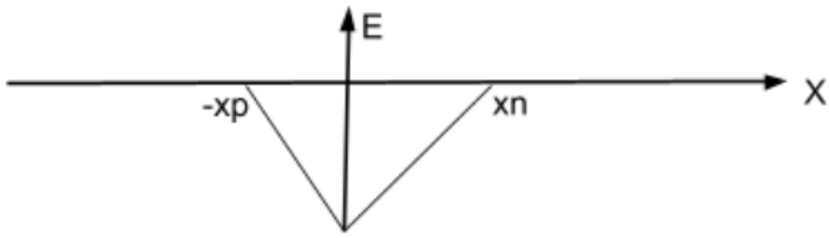
Question **8**

Correct

Mark 1.00 out of 1.00

(Negative mark of -0.25 will be awarded for wrong answer)

The given figure presents the position-dependent electric field profile of a forward-biased pn junction diode. If the slopes of the quasi-Fermi level of electron and hole are denoted as dE_{fn}/dx and dE_{fp}/dx, respectively,



Select the most suitable option.

Select one:

- Both dE_{fn}/dx and dE_{fp}/dx are positive throughout the so-called neutral and depletion regions ✔
- dE_{fn}/dx is negative and dE_{fp}/dx is positive throughout the so-called neutral and depletion regions
- dE_{fp}/dx is positive within the so-called neutral region and negative within the depletion region
- dE_{fn}/dx is negative within the so-called neutral region and positive within the depletion region

Your answer is correct.

The correct answer is: Both dE_{fn}/dx and dE_{fp}/dx are positive throughout the so-called neutral and depletion regions

Question **9**

Correct

Mark 1.00 out of 1.00

A silicon p^+n junction diode has doping concentration of $N_D = 1x10^{15}/cm^3$ on the n-side. For silicon, assume $\epsilon_S = 1x10^{-12}$ F/cm.

When the depletion layer width (W) = $1x10^{-3}$ cm, the maximum field strength is ..... x 10^4 V/cm.

Answer: 16 ✔

The correct answer is: 16

Question **10**

Correct

Mark 1.00 out of 1.00

$I = I_{SC} - I_0 e^{V/V_T}$

$I_{SC} = I_0 e^{V_{OC}/V_T}$

A solar cell has open circuit voltage V_{OC}=0.5 V and short circuit current I_{SC}=50 mA. Assume thermal voltage V_T=0.026 V. If another solar cell with V_{OC}=0.55V and I_{SC}=40 mA are connected in parallel,

The open-circuit voltage of this overall parallel cell combination will be (correct up to 3 digits after the decimal point) ..... V.

$V_{OC} = V_T \ln\left(\frac{I_{SC\,total}}{I_{0\,total}}\right)$

Answer: 0.5124 ✔

The correct answer is: 0.512

Question **11**

Correct

Mark 1.00 out of 1.00

An abrupt silicon $p^+/n/n^+$ junction diode is reverse biased with a voltage V_R. The built-in potential is V_{bi}. The n-layer width is $1x10^{-3}$ cm. The doping concentration in the n – layer is $5x10^{13}/cm^3$. For silicon, assume $\epsilon_S = 1x10^{-12}$ F/cm.

The maximum electric field in the junction when $V_R + V_{bi}$ = 64 V is ..... x 10^4 V/cm.

Answer: 6.8 ✔

The correct answer is: 6.8

Question **12**

Correct

Mark 1.00 out of 1.00

In a long abrupt p^+n diode of cross-sectional area (A) equal to 10^{-4} cm^2, when the applied forward bias (V_f) is 0.6V, the hole concentration at the edge of the depletion region in the n-region is $1x10^{14}/cm^3$. The hole diffusion constant (D_p) and lifetime (τ_p) in the n-region are 10 cm^2/sec and 1 μs respectively. Assume V_T = kT/q = 26 mV.

The diffusion (or stored charge) capacitance (C_d) at V_f = 0.6 V (in pF) is .....

Answer: 194.6 ✔

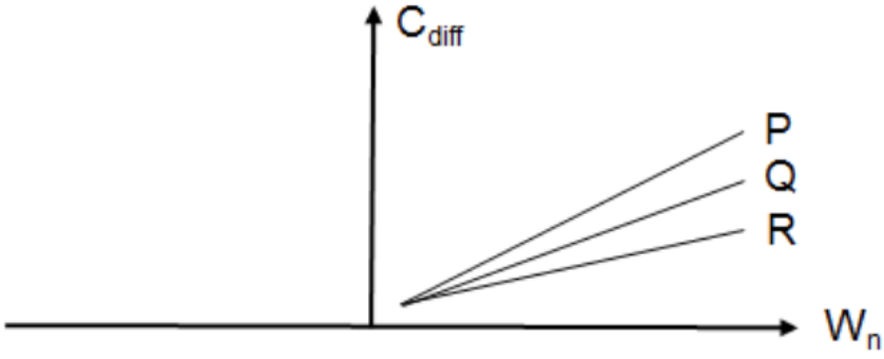
The correct answer is: 194

Question **13**

Correct

Mark 1.00 out of 1.00

(Negative mark of -0.25 will be awarded for wrong answer)



Three sets of abrupt p^+n junction diodes are fabricated with short n-regions ($W_n << L_p$) and they are referred to as P, Q and R in the given figure where neutral n-region widths (W_n) dependent diffusion capacitances (C_{diff}) at T=300 K are plotted. Except for the underlying semiconductor materials, these three sets of diodes are identical and are characterized at the same bias condition. If the energy bandgap for the set of diodes corresponding to P is E_{GP}=1 eV,

Select the most suitable option.

Select one:

- Energy bandgaps for the sets of diodes corresponding to Q and R are E_{GQ}=0.8 eV and E_{GR}=0.9 eV, respectively
- Energy bandgaps for the sets of diodes corresponding to Q and R are E_{GQ}=0.8 eV and E_{GR}=1.4 eV, respectively
- Energy bandgaps for the sets of diodes corresponding to Q and R are E_{GQ}=1.1 eV and E_{GR}=1.4 eV, respectively ✔
- Energy bandgaps for the sets of diodes corresponding to Q and R are E_{GQ}=1.4 eV and E_{GR}=1.1 eV, respectively

Your answer is correct.

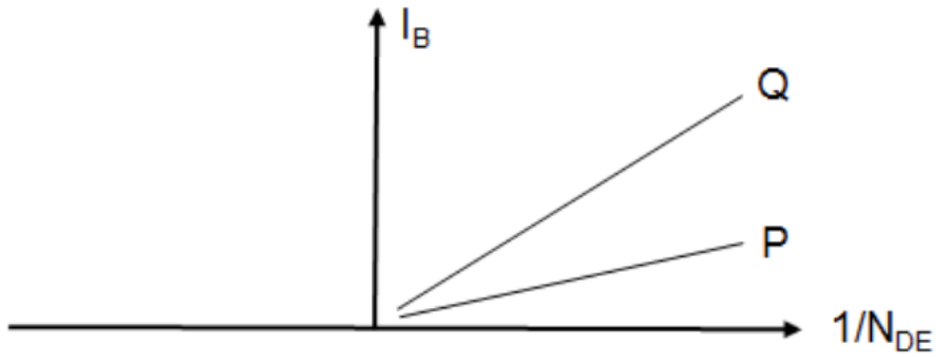
The correct answer is: Energy bandgaps for the sets of diodes corresponding to Q and R are E_{GQ}=1.1 eV and E_{GR}=1.4 eV, respectively

Question **14**

Correct

Mark 1.00 out of 1.00

(Negative mark of -0.25 will be awarded for wrong answer)



Two sets of npn transistors (all with short emitter, base and collector) are fabricated and they are referred to as P and Q in the given figure where $1/N_{DE}$ (N_{DE} being the doping concentration in the emitter region) versus base currents (I_B) at V_{BC}=0 V, V_{BE} = 0.6 V and T=300 K are plotted. For all the transistors in both sets, the recombination current within the base region (I_{Br}) can be assumed to be zero. Except for the underlying semiconductor materials and neutral emitter widths, these two sets of transistors are identical and are characterized at the same bias condition. If the neutral emitter width and energy bandgap of the material of transistors set Q are W_{nEQ}= 1.5 μm and E_{GQ}=1.5 eV, respectively,

Select the most suitable option.

Select one:

- Neutral emitter width and energy bandgap of the material of set P are W_{nEP}= 1.6 μm and E_{GP}=1.7 eV, respectively ✔
- Neutral emitter width and energy bandgap of the material of set P are W_{nEP}= 1.4 μm and E_{GP}=1.3 eV, respectively
- Neutral emitter width and energy bandgap of the material of set P are W_{nEP}= 1.2 μm and E_{GP}=1.2 eV, respectively
- Neutral emitter width and energy bandgap of the material of set P are W_{nEP}= 1.1 μm and E_{GP}=1.1 eV, respectively

Your answer is correct.

The correct answer is: Neutral emitter width and energy bandgap of the material of set P are W_{nEP}= 1.6 μm and E_{GP}=1.7 eV, respectively

Question **15**

Correct

Mark 1.00 out of 1.00

In a p-n junction of uniform cross-section, the width of the neutral n-region (W_n) is much less than the hole diffusion length (L_p) and the width of the neutral p-region (W_p) is much less than the electron diffusion length (L_n). For this device, $W_p = 2W_n$. Also, the resistivities of the p-region and n-region are ρ_p = 1 Ω-cm and ρ_n = 0.1 Ω-cm respectively.

If the device is forward biased so that the current flowing through the device is 6mA, the electron current component (I_n) (in mA) is ……

Answer: 5 ✔

The correct answer is: 5